



OPF Girls College F-8/2, Islamabad
FIRST TERM EXAMINATION AUGUST 2026
MATHEMATICS : M- II (Science Group)

Time allowed: 02:15 hours

Total Marks Sections B & C 40

SECTION - B (24 Marks)

Q. 2 Solve the following questions

(6×4 =24)

(i)	Simplify: $\frac{(81)^n \cdot 3^5 - (3)^{4n-1} (243)}{(9^{2n})(3^3)}$	4	OR	$\sqrt{\frac{(216)^{2/3} \times (25)^{1/2}}{(0.04)^{-3/2}}}$	4
(ii)	Use law of logarithm to evaluate $\log_2 5\sqrt{3}$	4	OR	Factorize the following: $4x^2 - y^2 - 2y - 1$	4
(iii)	Find the HCF of the following by factorization: $ax^2 + 2a^2x + a^3, 2ax^2 - 4a^2x - 6a^3, 3(ax + a^2)^2$	4	OR	Find the LCM of: $x^2 - 12xy + 16y^2, x^2 - 6xy + 8y^2$ and $3x^2 - 12y^2$	4
(iv)	Show that points A (-1, 2), B (7, 5) and C (-2, 6) are vertices of a right angled triangle.	4	OR	A(-2, 1) and B(6, 5) are opposite ends of the diameter of a circle. Find coordinates of its mid point.	4
(v)	First polynomial is $x^2 - x - 6$, HCF and LCM are $(x - 3)$ and $x^3 + 11x^2 + 12x - 12$ respectively. Find the other polynomial.	4	OR	Express the product $\frac{x^3-8}{x^2-4} \times \frac{x^2+6x+8}{x^2-2x+1}$ as an algebraic expression reduced to lowest form.	4
(vi)	Solve and Check: $\sqrt{\frac{z}{z+3}} = \sqrt{\frac{z+2}{z+6}}$	4	OR	Construct the following triangles. ΔXYZ when $XY = 5\text{cm}, \angle X = 30^\circ, \angle Z = 105^\circ$	4

SECTION - C (Marks 16)

Note: Attempt the following questions.

(2 × 8=16)

Q.3	Evaluate with the help of logarithm. $\frac{\sqrt{37.54} \times (38.4)^{1/2}}{\sqrt[3]{31.5}}$	8	OR	Simplify the following: $\frac{x+3}{x^2-3x+2} + \frac{x+2}{x^2-4x+3} + \frac{x+1}{x^2-5x+6}$	8
Q.4	Volume of a cubical container is given by the expression: $(x^3 - 6x^2y + 12xy^2 - 8y^3)m^3$ Find: (a) Expression for length of each side (b) Expression for area of base (c) Expression for total surface area (d) Expression for cost of painting all outer surfaces at the rate of Rs. 50/m ²	8	OR	Find the square root of the following by division method. $\frac{x^2}{y^2} - 10\frac{x}{y} + 27 - 10\frac{y}{x} + \frac{y^2}{x^2}$ $(x \neq 0, y \neq 0)$	8